

Product Guide

NHD-128-NDI-TRX

NDI-NetworkHD Bi-directional Gateway



WyreStorm

**Award-Winning Audio-Visual
Solutions Manufacturer**

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Product Overview



NDI-NetworkHD Bi-directional Gateway

The NHD-128-NDI-TRX is a powerful and versatile gateway designed to enable bidirectional conversion between NDI and NetworkHD streams. When paired with NetworkHD 100 Series products, it provides smooth and efficient integration within IP-based video workflows.

With support for up to 8 channels of NDI-to-NetworkHD conversion and 1 channel of NetworkHD-to-NDI output, the NHD-128-NDI-TRX offers a scalable and flexible solution for managing high-density video sources. It is especially well-suited for multi-camera environments where reliable stream conversion is essential.

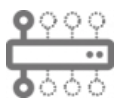
Ideal for applications such as digital courtrooms, lecture halls, and other AV-over-IP scenarios, this gateway ensures dependable interoperability between systems while streamlining deployment and reducing infrastructure complexity.

Main Features



Reliable NDI Stream to NetworkHD Stream Conversion

Convert NDI streams to NetworkHD streams, enabling seamless routing to decoders. Ideal for digital courtrooms, lecture halls, and other multi-camera AV-over-IP systems.



Seamless NetworkHD Stream to NDI Stream Conversion

Effortlessly convert sources from NetworkHD into NDI streams—ideal for AV-over-IP environments. Enables cost-effective migration to IP workflows without the need to replace legacy devices.



Crystal-Clear 4K@30Hz Video Quality

Supports 4K@30Hz resolution for both input and output, ensuring ultra-clear visuals for presentations, signage, and live video. Delivers exceptional image detail with stable signal transmission.



Video Preview Supported

Supports preview streaming for real-time video monitoring—perfect for validating sources during setup or managing large deployments.



Intuitive Web UI for Easy Control

Equipped with a powerful and user-friendly Web UI for configuration, monitoring, and firmware updates. Simplifies setup for installers and boosts productivity in dynamic environments.



Comprehensive API and Protocol Support

Offers versatile API interfaces supporting multiple control protocols (Telnet, SSH, etc.), ideal for system integrators building customized control platforms for enterprise or broadcast needs.



Device Information Monitoring with LCD Screen

The front-panel LCD displays IP address and firmware version, providing quick check for device basic information.

Why Choose Us: 1. Bring More Possibility to AV over IP System

Breaks the barrier between NetworkHD and NDI networks, enabling seamless interoperability between two systems through conversion.

Bi-Directional Conversion

Supports both NDI stream to NetworkHD stream and NetworkHD stream to NDI stream conversion at the same time, enabling efficient AV signal management through one device.

High-Quality Performance

Deliver 4K@30Hz video and low latency transmission for professional-grade visuals and real-time interaction across networked environments.

Bring Multiview to NDI

Supports the conversion of multiview output from the NHD-150-RX into an NDI stream, allowing multiple video sources to be seamlessly integrated into any NDI-based system.



Powerful · Reliable · Simple

Why Choose Us: 2. Easy Deployment and Effortless Control

Enable seamless workflows with one device that converts between NDI and NetworkHD streams simultaneously – reducing hardware requirements and increasing integration flexibility.

Effortless Connectivity

Streamline installation with just a single network cable that carries power, video, and control in one connection. Reduce clutter and minimize setup time.

Seamless Integration

Seamlessly integrates into existing AV infrastructures with native support for both NDI and NetworkHD, making it an ideal choice for lecture halls, conference rooms, and digital courtrooms.

Boost Operational Efficiency

Enable fast setup and centralized control via intuitive Web UI and APIs—freeing up time for content creation, not system maintenance.

Real-Time LCD Status Monitoring

Front-panel display shows IP address and firmware version in real time, providing quick check for device information.

Interface of the Product

Front Panel



Featuring:

LCD Screen

The LCD screen on the front panel displays the device's IP address and firmware version in real time.

All-in-One LAN Port

Support PoE on the single LAN port. Enable bidirectional conversion between NetworkHD and NDI streams.

Rear Panel



Specifications & Packing List

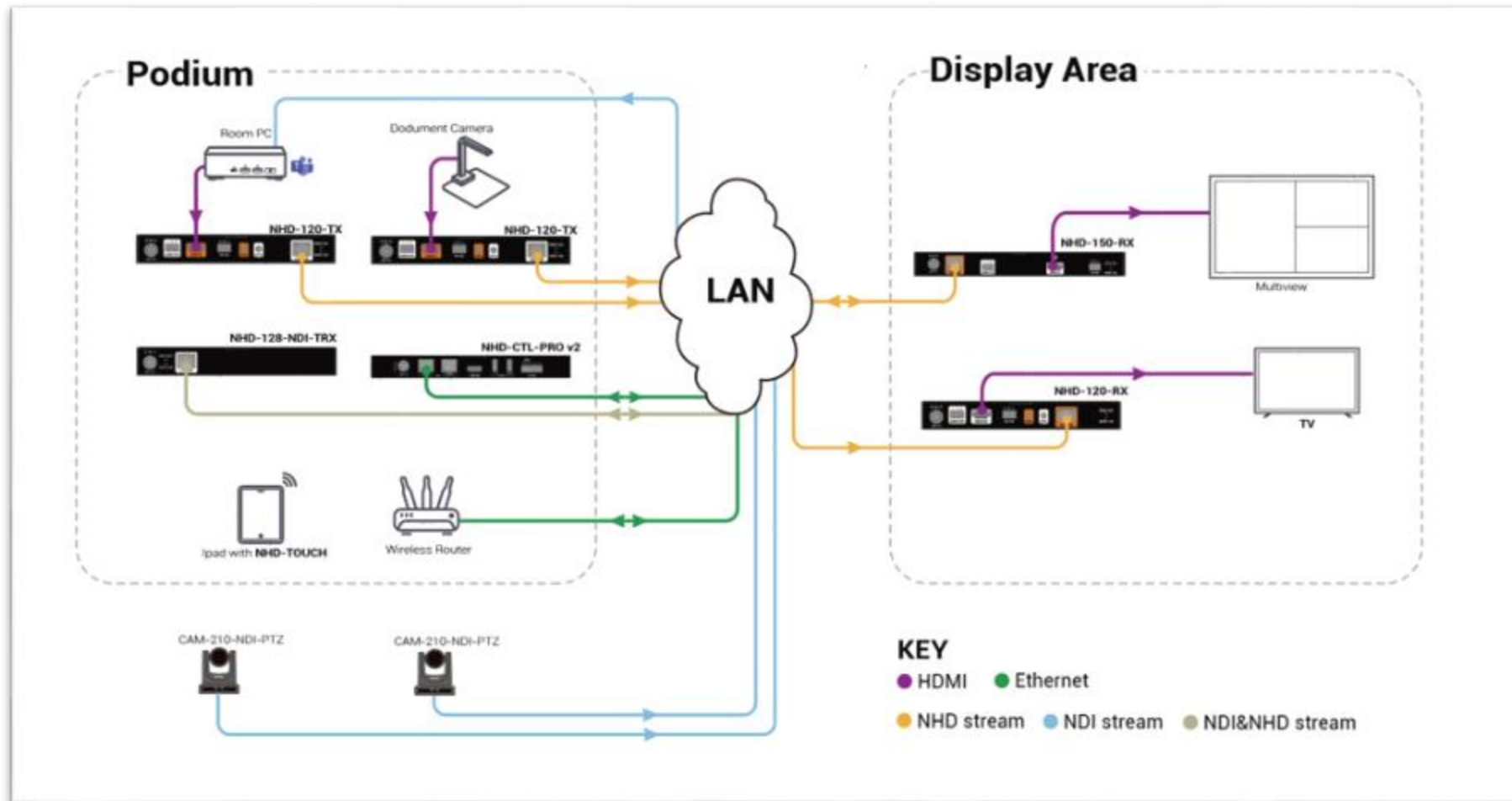
Main Specifications		
Input Video	NetworkHD	Timing up to 4K@30Hz Bitrate up to 16Mbps
	NDI	Main stream: Timing up to 4K@30Hz Bitrate up to 16Mbps Preview stream: Timing up to 480P@60Hz Bitrate up to 1Mbps
Output Video	NetworkHD	Timing up to 4K@30Hz Bitrate up to 16Mbps
	NDI	Main stream: Timing up to 4K@30Hz Bitrate up to 20Mbps Preview stream: CIF@30 Bitrate up to 512Kbps
NDI Protocol		NDI HX (NDI 6)
LAN		1x LAN: 8-pin RJ-45 Female 10/100/1000Base-T PoE 802.3af
Control Methods		Web UI Telnet Telnet TLS SSH

What's in the Box?

- 1x NHD-128-NDI-TRX
- 4x Wall Plastic Mounting Ears
- 2x Rack Mounting Ears for NHD-000-RACK4 vertical installation.

Typical Connection & Applications

The NHD-128-NDI-TRX supports bi-directional conversion between NetworkHD and NDI streams. It is ideal for digital courtrooms, lecture halls, and other environments requiring seamless integration between the two systems. Please refer to the typical connection diagram below.



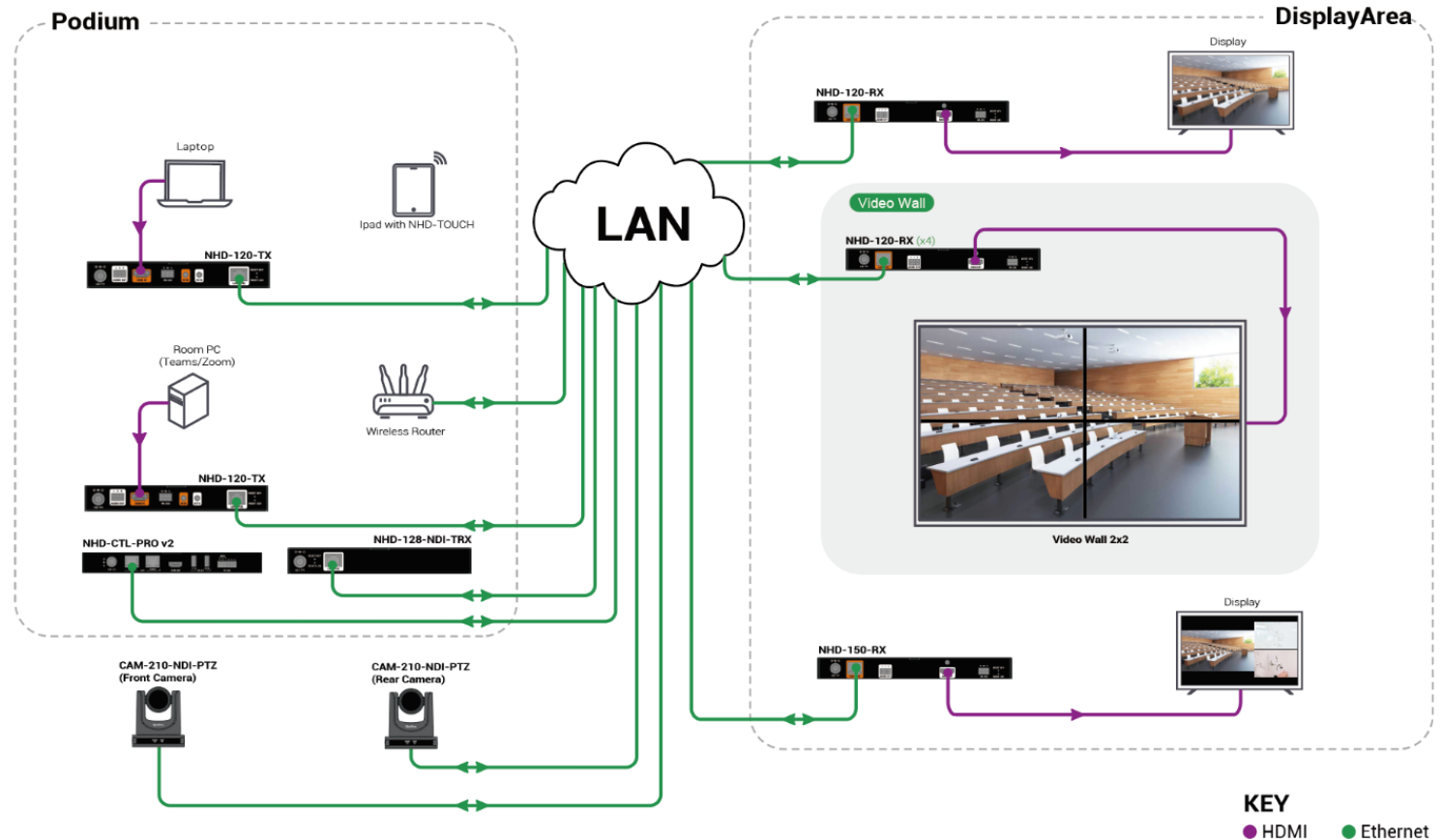
Typical Connection & Applications

Application Scenario 1: Lecture Hall

Deliver a comprehensive remote teaching solution that connects in-class and online students. Both groups can simultaneously view shared courseware and live camera feeds of instructors and classmates, creating an engaging and immersive learning experience across locations.

Key Benefits Include:

- **Multiview Function:** Present camera feeds and course content on the same screen to provide a complete view of the lecture environment.
- **NDI Integration:** Easily incorporate NDI cameras for fast setup and flexible streaming. NDI signals can also be converted to NetworkHD format for local display on monitors.
- **Remote Teaching:** Stream NDI camera feeds directly into conferencing platforms like Zoom or Microsoft Teams, enabling real-time classroom access for remote students. Perfect for hybrid teaching, distance learning, and modern educational models.



Typical Connection & Applications

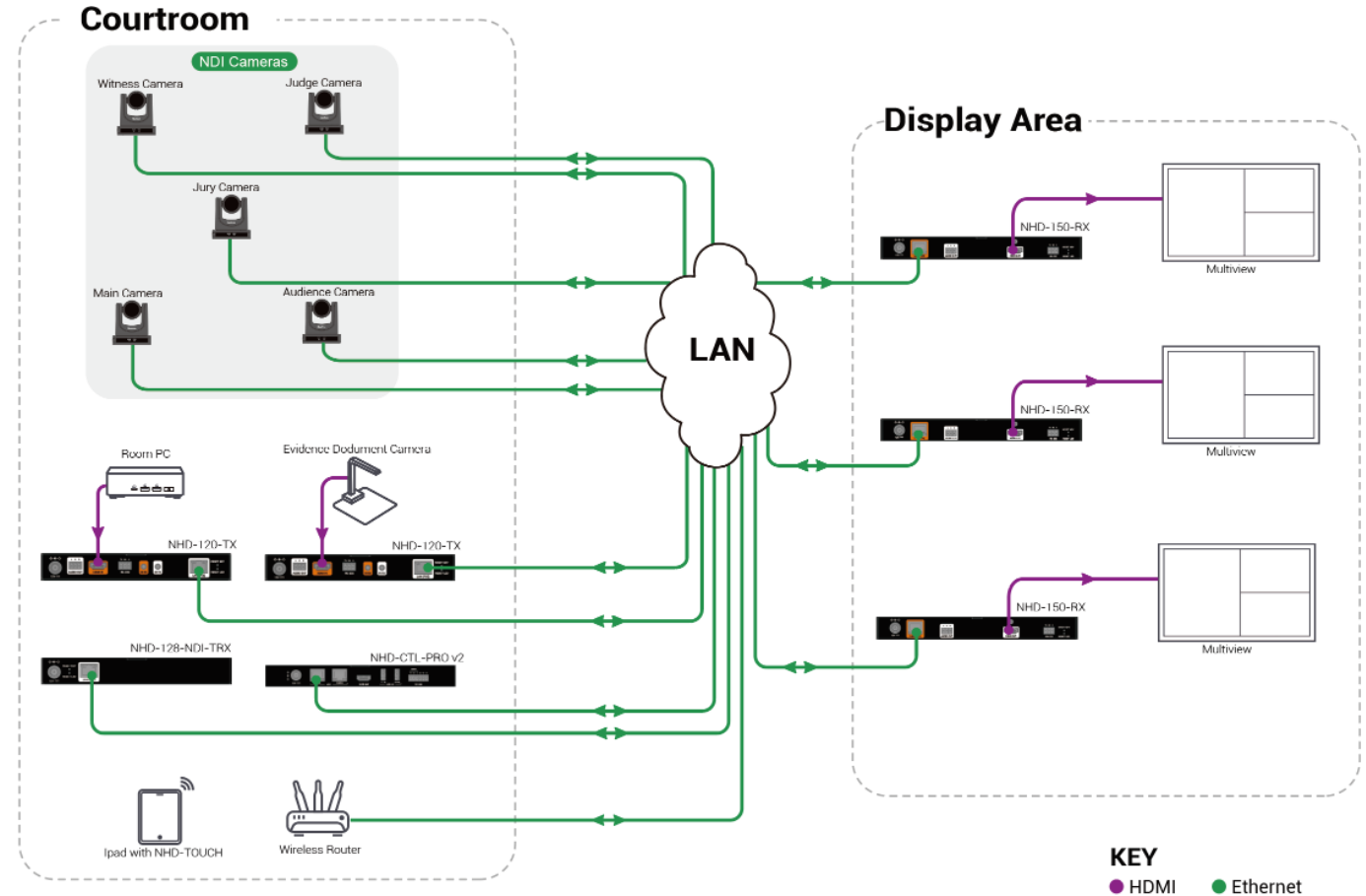
Application Scenario 2: Digital Courtroom

This solution delivers a comprehensive AV system tailored for a digital courtroom environment.

It enables real-time display of multiple camera feeds and shared evidence documents in a multiview layout. The combined multiview output can also be converted to NDI for remote viewing or live streaming.

Key Benefits Include:

- **Multiview Function:** Simultaneous display of camera feeds and shared content for a complete courtroom overview.
- **NDI Integration:** Supports NDI camera convert into NetworkHD stream for AV over IP system.
- **Full-Court Coverage:** Comprehensive video input from multiple courtroom cameras, including judge, jury, and evidence views.
- **Centralized Control:** Intuitive system control via iPad with NHD-TOUCH and NHD-CTL-PRO for streamlined operation.
- **Flexible Signal Distribution:** Seamlessly integrates both NetworkHD and NDI protocols over a standard LAN, ensuring reliable and scalable AV transmission.



Comparison with Other Competitors

		WyreStorm NHD-128-NDI-TRX	UrayCoder UHE265-4-4K	Kiloview N40	Magewell 641100000	Magewell 640100000
	Conversion Type	NetworkHD to NDI & NDI to NetworkHD	HDMI to NDI	HDMI to NDI & NDI to HDMI	NDI to HDMI	HDMI to NDI
Input	Video Input	8x NDI stream, 1x NetworkHD stream	4x HDMI input	1x HDMI in, 1x NDI stream	1x NDI stream	1x HDMI in
	Input Timing	Up to 4K@30Hz NetworkHD stream Up to 4K@30Hz NDI stream	Up to 4K@30Hz HDMI input	Up to 4K@60Hz NDI stream Up to 4K@60Hz HDMI input	Up to 4K@60Hz NDI stream	Up to 4K@60Hz HDMI input
Output	Video Output	8x NetworkHD stream, 1x NDI stream	4x NDI stream	1x HDMI out, 1x NDI stream	1x HDMI out	1x HDMI loop out 1x NDI stream
	Output Timing	Up to 4K@30Hz NDI stream Up to 4K@30Hz NetworkHD stream	Up to 4K@30Hz NDI stream	Up to 4K@60Hz HMDI output Up to 4K@60Hz NDI stream	Up to 4K@60Hz HDMI output	Up to 4K@60Hz NDI stream
Others	NDI® Codec Capability	NDI® HX(NDI 6)	N/A	NDI High Bandwidth	NDI® HX/NDI High Bandwidth(NDI 5.x)	NDI® HX/NDI High Bandwidth(NDI 5.x)
	PoE	✓	×	✓	✓	✓
	LCD Screen	✓	×	×	×	×

Quick Installation Guide

Effortlessly set up your NHD-128-NDI-TRX with simple steps, ensuring seamless configuration and an exceptional viewing experience.

- **Connect Interfaces:** Connect the CAT cable to a PoE network switch (or using PSU to power on the device) for NHD-128-NDI-TRX.
- **Start the Device:** Check the IP Address from the LCD screen, then log into the Web-UI for easy device configuration.
- **Create Conversion:** Create your specified configuration via Web-UI. For detailed instructions, refer to the user manual.
- **Enjoy NDI and NetworkHD Conversion:** NDI streams and NetworkHD streams can be freely converted and flexibly configured within the AV-over-IP system.

Important Notes for Installation

- Check your network settings and confirm that the device is on the correct subnet.
- Double-check configuration settings in the Web-UI to optimize functionality.
- For complex settings, refer to the documentation or seek professional assistance if necessary.

Introduction of Web Configuration Interface

The Web UI of the NHD-128-NDI-TRX is designed for intuitive device management and control through any web browser. With the Web UI, you can easily configure and manage your conversion setup with the following features:

Key Web UI Configurations:

- **Conversion Configuration:** Setting the IP address and conversion status between NDI and NetworkHD stream.
- **Conversion Management:** Activate/Deactivate each conversion channel.

Final Thoughts

Q & A Session

Got questions? Don't hesitate to ask away.

